

Assignment 1: Poisson–Gamma Posterior

CSDS 611 — Advanced Topics and Methods of Data Science I

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Model and parameterization

Consider the Bayesian model

$$w_n \mid \lambda \sim \text{Poisson}(\lambda), \quad n = 1, \dots, N, \quad \lambda \sim \text{Gamma}(\phi, \psi).$$

Assume that the observations are conditionally independent given λ , with $w_n \in \{0, 1, 2, \dots\}$ and $\phi, \psi > 0$. Gamma distribution is parameterized by shape ϕ and scale ψ . Let $\mathbf{w} = (w_1, \dots, w_N)$ and $S = \sum_{n=1}^N w_n$. We seek the posterior density $p(\lambda \mid \mathbf{w})$.

1. Likelihood

The Poisson probability mass function is

$$p(w_n \mid \lambda) = \frac{\lambda^{w_n} e^{-\lambda}}{w_n!}.$$

Conditional independence gives the joint likelihood

$$\begin{aligned} p(\mathbf{w} \mid \lambda) &= \prod_{n=1}^N \frac{\lambda^{w_n} e^{-\lambda}}{w_n!} \\ &= \frac{\lambda^S e^{-N\lambda}}{\prod_{n=1}^N w_n!}. \end{aligned}$$

2. Prior

The Gamma prior has density

$$p(\lambda) = \frac{1}{\Gamma(\phi)\psi^\phi} \lambda^{\phi-1} e^{-\lambda/\psi}, \quad \lambda > 0.$$

3. Apply Bayes' rule

By Bayes' rule,

$$\begin{aligned}
 p(\lambda \mid \mathbf{w}) &\propto p(\mathbf{w} \mid \lambda) p(\lambda) \\
 &= \frac{\lambda^S e^{-N\lambda}}{\prod_{n=1}^N w_n!} \frac{\lambda^{\phi-1} e^{-\lambda/\psi}}{\Gamma(\phi) \psi^\phi} \\
 &\propto \lambda^{\phi+S-1} \exp\left[-\left(N + \frac{1}{\psi}\right) \lambda\right].
 \end{aligned}$$

The final proportionality omits only factors that do not depend on λ .

4. Normalize and identify the posterior

Introduce a normalizing constant C and write

$$p(\lambda \mid \mathbf{w}) = C \lambda^{\phi+S-1} \exp\left[-\left(N + \frac{1}{\psi}\right) \lambda\right], \quad \lambda > 0.$$

The posterior must integrate to one. Using the substitution $t = (N + 1/\psi)\lambda$, we obtain

$$\begin{aligned}
 1 &= \int_0^\infty p(\lambda \mid \mathbf{w}) d\lambda \\
 &= C \int_0^\infty \lambda^{\phi+S-1} e^{-(N+1/\psi)\lambda} d\lambda \\
 &= C \left(N + \frac{1}{\psi}\right)^{-(\phi+S)} \int_0^\infty t^{\phi+S-1} e^{-t} dt.
 \end{aligned}$$

The remaining integral equals $\Gamma(\phi + S)$, by normalization of the Gamma density with shape $\phi + S$ and scale 1. Therefore,

$$1 = C \frac{\Gamma(\phi + S)}{\left(N + \frac{1}{\psi}\right)^{\phi+S}}, \quad C = \frac{\left(N + \frac{1}{\psi}\right)^{\phi+S}}{\Gamma(\phi + S)}.$$

Substituting C gives the normalized posterior density:

$$p(\lambda \mid \mathbf{w}) = \frac{\left(N + \frac{1}{\psi}\right)^{\phi+S}}{\Gamma(\phi + S)} \lambda^{\phi+S-1} \exp\left[-\left(N + \frac{1}{\psi}\right) \lambda\right], \quad \lambda > 0.$$

Comparing this expression with the Gamma density gives the updated shape and scale:

$$\phi' = \phi + \sum_{n=1}^N w_n, \quad \psi' = \frac{1}{N + 1/\psi} = \frac{\psi}{1 + N\psi}.$$

Thus,

$$\lambda \mid w_1, \dots, w_N \sim \text{Gamma}\left(\phi + \sum_{n=1}^N w_n, \frac{\psi}{1 + N\psi}\right) \quad (\text{shape--scale}).$$

The posterior remains in the Gamma family, so the Gamma prior is conjugate to the Poisson likelihood.